Week 2 lecture 2

Linear algebra + MATLAB Syntax

Recall

Recall

- Vectors
  - Row, Column
- Matrices

Vector Operations

- Addition $\vec{u} + \vec{v} = [u_1 + v_1, u_2 + v_2, \ldots]$
- Subtraction $\vec{u} - \vec{v} = [u_1 - v_1, u_2 - v_2, \ldots]$
- Scaling $\vec{u}\alpha = [u_1\alpha, u_2\alpha, \ldots]$
- Dot Product $\vec{u} * \vec{v} = u_1v_1 + u_2v_2 + \ldots$
- Length $|\vec{u}| = \sqrt{u_1^2 + u_2^2 + \ldots}$

Matrix Arithmetic

- Addition
- Subtraction
- Multiplication
- Traspose
- Inverse

Matrix indexing

- <row> x <col>
- Only 2D matrices

Reduced Row echelon form

- System of equations

$$1x_1 + 2x_2 = 5$$
$$3x_1 + 9x_2 = 21$$

- Matrix

$$\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 3 & 9 & 21 \end{array}\right]$$

- Solve

$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right]$$

Guass-Jordan Elimination

- Algorithm for solving linear systems of equations

Matrix Algebra

- Just like algebra… but with matrices

MATLAB Syntax

MATLAB Vector

Syntax for creation

Syntax for indexing

Indexing returns value

```
rv = [1,2,3]
cv = [1;2;3]
M  = [rv;rv;rv]
M(1,2)
squid = M(3,1) + M(1,1)
```

LAB

- LAB today is homework due before next class